

Hillol Kumar Barman

Software Engineer | Full-Stack

[GitHub](#) | [Portfolio](#) | [LinkedIn](#)

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PROFESSIONAL SUMMARY

Full-stack engineer with a Master's in Software Engineering (Distinction, UOW) and hands-on experience building production-ready web applications using React, TypeScript and Python/FastAPI. Comfortable across the stack — from designing REST APIs and applying OOP fundamentals on the backend, through to building responsive React/HTML/CSS interfaces and deploying containerised services on AWS. Looking to bring that end-to-end capability to a team building high-traffic digital products.

EDUCATION

Master of Computer Science (Software Engineering) – Distinction

University of Wollongong, NSW | July 2023 – June 2025

- Achieved 5 High Distinctions and 6 Distinctions throughout the masters program
- Lead developer in capstone project

Bachelor of Computer Science and Engineering

Institute of Engineering and Management (IEM), Kolkata, India | CGPA: 3.50 / 4.00

- Acted as Technical Head in major tech events

PROJECTS

Github Visualiser ([GitHub](#) | [Live](#)):

- Built a full-stack web application using **React**, **TypeScript** and **Tailwind CSS** to visualise GitHub repository commit history as an interactive graph
- Implemented **GitHub OAuth** via **Supabase** with protected API routes and PostgreSQL-backed session storage
- Designed a **REST API** in **Python/FastAPI** using object-oriented service classes and **Pydantic** models to proxy GitHub REST API calls with a server-side token, keeping credentials out of the browser
- Wrote unit tests for backend API routes using **pytest** to validate authentication and request-handling logic
- Containerised both services with Docker, orchestrated with **Docker Compose**, and deployed to **AWS EC2** with an automated one-command redeploy script for fast iteration

Grounded - AI Knowledge Base Chatbot - Beta ([GitHub](#)):

- Built a full-stack platform enabling users to upload documents and chat with their content using **Retrieval-Augmented Generation (RAG)**
- Designed an ingestion pipeline supporting PDF, DOCX, CSV, XLSX, URLs, and ZIP archives, with token-aware chunking via **tiktoken** and batch embedding via **OpenAI API**
- Implemented semantic search using **Supabase pgvector** and streamed LLM responses token-by-token over Server-Sent Events using the **Anthropic Claude API**
- Authenticated users via Supabase Auth with **JWT validation** on all API routes and automatic token refresh; structured the **FastAPI** backend around object-oriented service and repository classes

- Containerised both services with **Docker** and orchestrated with **Docker Compose**

Personal Portfolio Website with Integrated Code Editor ([GitHub](#) | [Live](#)):

- Developed a full-stack web platform using **React**, **JavaScript/TypeScript**, **HTML**, **CSS**, **FastAPI** and **Supabase** (PostgreSQL), taking the project from design through to production deployment
- Built an in-browser **Monaco-based** code editor with reusable React components for UI consistency, and styled responsive layouts using **Tailwind CSS**
- Designed **REST APIs** using **FastAPI** for authentication, user data management, and community code-sharing features, with SQL queries against a **PostgreSQL** database
- Integrated **Stripe** for user payments and support features, including webhook handling on the **FastAPI** backend
- Containerized backend services using **Docker** for scalable deployment

EXPERIENCE

Freelance UI Designer & Developer

Drivers4Me (Startup), Kolkata, India

Dec 2020 – Feb 2023

- Developed and maintained **RESTful APIs** in **Python/Flask** supporting production application features used in real-world customer workflows
- Built reusable frontend components with **React**, **JavaScript/TypeScript**, **HTML** and **CSS** using **Next.js**, applying object-oriented and component-based design to improve UI consistency and development efficiency
- Collaborated directly with founders and stakeholders to gather requirements, deliver features and respond to issues in production, working in a fast-paced environment
- Created **UI/UX** design components using **Figma** and **Adobe Illustrator**
- Worked across the full development lifecycle — from requirements discussions with the founders through implementation, testing and release — in a small, fast-moving team

Console Operator

NPG Urbanista Convenience Store, Albion Park NSW

Mar 2025 – Present

- Delivered high-volume customer service in a fast-paced retail environment
- Managed POS systems and performed accurate cash reconciliation
- Demonstrated reliability and independent decision-making during solo shifts

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, HTML, CSS, SQL

Frontend: React, Next.js, Tailwind CSS, Responsive Web Design

Backend: Node.js (Next.js), FastAPI, Flask, REST API Design

Testing: pytest, unit & integration testing

Databases: PostgreSQL, MySQL

Tools & Platforms: Docker, Docker Compose, AWS EC2, Git, CI/CD basics, Supabase, Stripe, Vercel, Railway

Concepts: Object-Oriented Programming, REST API Design, OAuth & JWT Authentication, Full SDLC, Scalable System Design

REFERENCES

Available upon request.